

Project Proposal: AI-Enabled Two-Tier Bovine Mastitis Early Warning System

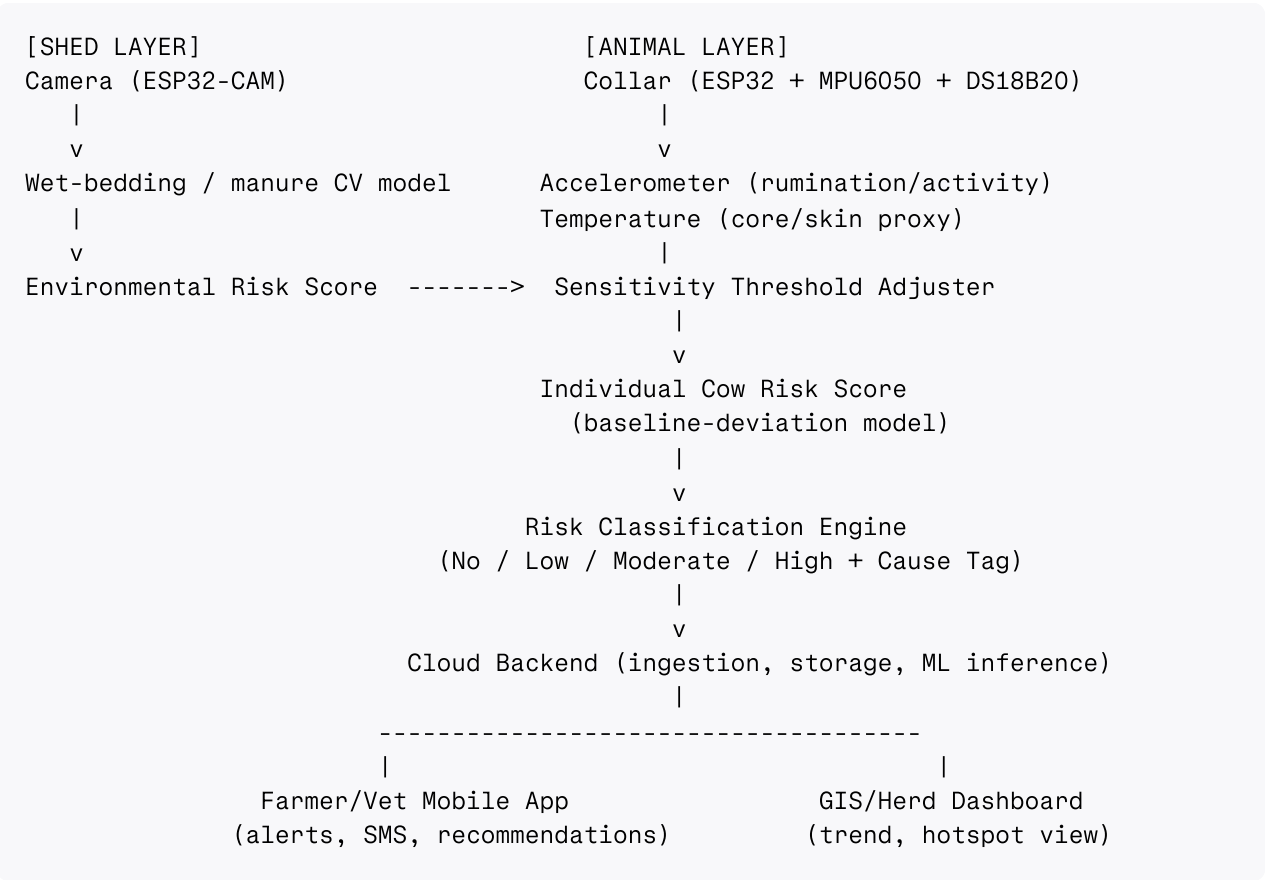
1. Project Summary

A low-cost, dual-layer predictive system for early detection of bovine mastitis, combining:

- **Herd/shed-level environmental monitoring** via a fixed camera + ambient sensors, continuously streaming data to flag rising *exposure risk* (wet bedding, poor manure management, humidity).
- **Individual cow-level monitoring** via a collar (accelerometer + temperature) that confirms *infection onset* through deviation from each animal's own rolling baseline.

The two layers work together: shed-level risk raises or lowers the alert sensitivity for individual cow monitoring, and individual cow signals confirm whether elevated exposure has actually turned into disease. This mirrors the real epidemiological split between environmental mastitis (housing/hygiene-driven) and contagious mastitis (cow-to-cow, milking-hygiene-driven), which the literature treats as two genuinely distinct causal pathways requiring different interventions.

2. System Architecture



3. Hardware Components

3.1 Individual Cow Collar

Component	Purpose	Approx. Cost (INR)
ESP32 (Dev Board)	Microcontroller + WiFi/BLE hub for collar	₹250–350
MPU6050 (accelerometer + gyro)	Activity, lying/standing time, rumination-proxy motion patterns	₹150–200
DS18B20 (waterproof temp probe)	Skin-contact temperature tracking, fever-spike detection	₹50–110
SX1278 LoRa module (433MHz)	Long-range, low-power transmission from pasture/shed to gateway	₹160–300
Small Li-ion battery + TP4056 charge module	Power source	₹150–250
6V mini solar panel (~100mA)	Trickle-charging for continuous field deployment	₹35–200
Weatherproof collar housing	Physical protection, animal safety	₹150–300 (custom/3D printed)
Estimated per-collar cost		~₹800–1,400

This is deliberately cheaper than commercial reticulorumen bolus systems (int'l products run well above ₹5,000/unit) because the collar approach avoids the invasive bolus form factor and its higher per-unit electronics cost, at the tradeoff of measuring skin/ambient temperature rather than true core body temperature — an explicit and defensible design tradeoff to state to judges.

3.2 Shed-Level Environmental Camera Node

Component	Purpose	Approx. Cost (INR)
ESP32-CAM (OV2640)	Captures periodic images of bedding/floor area	₹500–700
Mounting bracket + weatherproof enclosure	Fixed overhead placement	₹200–400
SX1278 LoRa module or WiFi (if shed has connectivity)	Data transmission to gateway	₹160–300
6V solar panel + battery pack	Continuous power in the shed	₹300–500
Estimated per-camera-node cost		~₹1,200–1,900

One camera node can realistically cover one pen/shed section, so cost scales with farm layout, not per animal — this is the key cost-efficiency argument versus per-animal sensors.

3.3 Gateway / Communication Hub

- One LoRa gateway (Raspberry Pi + LoRa concentrator, or a single ESP32 LoRa receiver for a hackathon-scale demo) per farm, aggregating all collar and camera nodes and forwarding to the cloud over WiFi/GSM/NB-IoT.
- Rationale for LoRa: long range (up to a few km line-of-sight) and very low power draw, well-suited to rural Indian farms with unreliable connectivity — collars/cameras don't need direct internet access, only the single gateway does.

Total estimated hardware cost for a small pilot (5 cows + 1 shed camera + 1 gateway): roughly ₹6,000–9,000, which is realistic to prototype within a hackathon budget and demonstrably cheaper than any commercial precision-livestock offering.

4. Software Architecture

4.1 Data Ingestion Layer

- Collars and camera node stream readings (accelerometer vectors, temperature, images/image-derived features) over LoRa to the gateway.
- Gateway batches and forwards to cloud via MQTT or HTTP REST to a message queue (AWS IoT Core, or a simple Lambda-fronted API Gateway endpoint given your AWS familiarity).
- All raw readings are timestamped and geo/animal-tagged, then written to a time-series-friendly store (DynamoDB with TTL for raw data, or a managed time-series DB if available).

4.2 Environmental Risk Model (Shed Layer)

- A lightweight CV model (transfer-learned on a small labeled image set — reuse your YOLO/segmentation experience) classifies each camera frame for:
 - Wet bedding area percentage (color/texture segmentation)
 - Standing water presence
 - Manure buildup/cleanliness score
- These three sub-scores combine into a single **Environmental Risk Score (0-100)** per shed, updated on each frame interval (e.g., every 15-30 min).
- This score feeds a rules-based **sensitivity adjuster**: high environmental risk lowers the anomaly threshold for the individual-cow model (more sensitive, catches subtler signals sooner); low risk raises the threshold (fewer false alerts).

4.3 Individual Cow Risk Model (Animal Layer)

- Each cow's collar data (activity level, lying time, rumination-proxy motion, temperature) is converted into a **rolling personal baseline** (e.g., 7-day trailing average) rather than a fixed population cutoff, since individual cows vary naturally.
- A time-series anomaly/trend model (start simple: rolling z-score deviation; upgrade to XGBoost or LSTM if time allows) flags sustained deviation from baseline as early risk, not

single-point threshold crossing — this is what published sensor studies show is necessary to achieve a genuine multi-day lead time rather than same-day detection.

- Output: a per-cow **Risk Score + Risk Category** (No/Low/Moderate/High Risk).

4.4 Cause Classification Layer

- When a cow crosses into Moderate/High risk, the system checks correlated shed-level Environmental Risk Score at the same time window:
 - High individual risk + high shed environmental risk → tag as **likely environmental** → recommend hygiene/bedding/housing correction.
 - High individual risk + low shed environmental risk → tag as **likely contagious** → recommend testing (SCC/California Mastitis Test) and isolating the animal, and flag milking-equipment hygiene.
- This cause tag is what drives differentiated, actionable recommendations rather than a generic "risk detected" alert.

4.5 Alerting & Notification Module

- Risk Category changes trigger push notification (mobile app) and SMS fallback (via AWS SNS or a Twilio-equivalent) to the farmer and assigned vet, in the farmer's chosen language.
- Alert content includes: animal ID, risk level, probable cause tag, and one concrete recommended action.

4.6 Dashboard & Visualization

- Farmer-facing mobile app: simple traffic-light view per animal, plus one-tap "what to do" recommendation card.
- Vet/cooperative-facing web dashboard: herd-level trend charts, environmental risk heatmap of the shed, and historical case log — this satisfies the GIS/hotspot visualization component of the original brief at farm scale (multiple sheds = multiple hotspots on one map).

4.7 Continuous Learning Loop

- Every confirmed clinical case (vet-entered outcome) is fed back as a labeled training example, incrementally improving the individual-cow and environmental models — directly satisfying the "continuously improving prediction accuracy" requirement from the original problem statement.

5. Data Flow Summary

1. Camera + collar sensors → LoRa → Gateway → Cloud ingestion.
2. Cloud computes Environmental Risk Score (CV model) and Individual Risk Score (time-series baseline-deviation model) in parallel.
3. Cause Classification Layer cross-references both scores.
4. Risk Classification Engine outputs final per-cow status + recommendation.
5. Alerts pushed to app/SMS; all data logged to dashboard for trend visualization.

6. Why This Design Is Defensible

- **Two distinct, honestly-framed time horizons:** environmental monitoring gives continuous, real-time exposure-risk alerting (satisfies "real-time" and "continuous monitoring" requirements); individual-cow monitoring gives a realistic 4-9 day lead time on actual infection onset, consistent with published sensor-based studies rather than an unsubstantiated flat 14-day claim.
- **Cost efficiency:** shed-level sensing scales per-pen, not per-animal, keeping hardware cost low relative to fully per-cow sensor suites.
- **Cause-specific recommendations:** separates environmental fixes from contagious-disease isolation actions, directly supporting antimicrobial-resistance mitigation goals by avoiding blanket treatment.
- **Reuses your existing CV/YOLO skillset** for the bedding/manure vision model, making this genuinely buildable within a hackathon timeframe rather than requiring unfamiliar tooling.